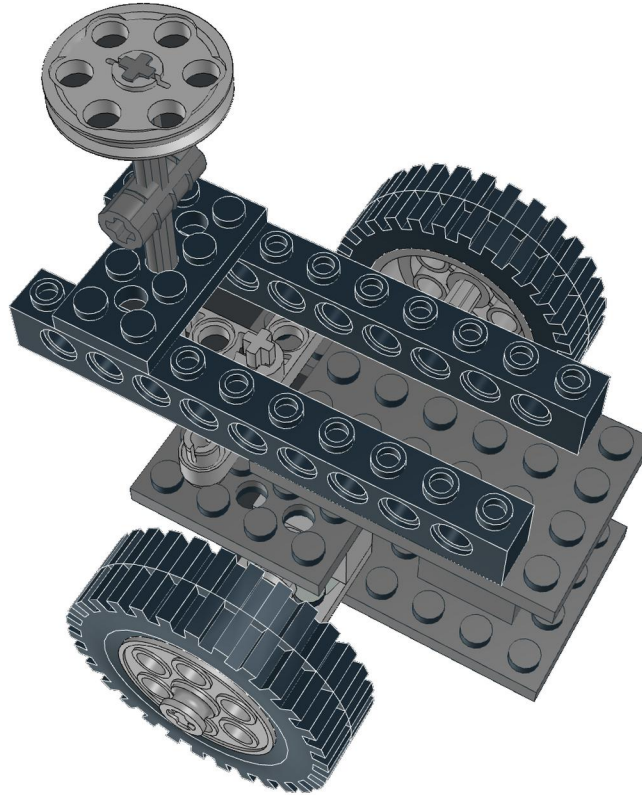


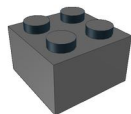
Bj and the Bear

Front axle test module · steering prototype

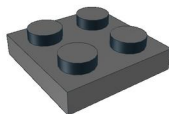


35 parts · 12 steps

Parts



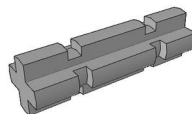
2x
3003 Dark Bluish Gray
Brick 2 x 2



1x
3022 Dark Bluish Gray
Plate 2 x 2



2x
3032 Dark Bluish Gray
Plate 4 x 6



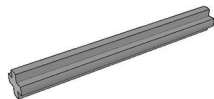
1x
32062 Light Bluish Gray
Technic Axle 2 Notched



1x
4519 Dark Bluish Gray
Technic Axle 3



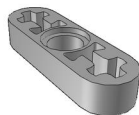
1x
3705 Dark Bluish Gray
Technic Axle 4



2x
32073 Light Bluish Gray
Technic Axle 5



1x
6538a Dark Bluish Gray
Technic Axle Joiner



1x
6632 Light Bluish Gray
Technic Beam 3 x 0.5 Liftarm



1x
32017 Light Bluish Gray
Technic Beam 5 x 0.5



2x
3700 Light Bluish Gray
Technic Brick 1 x 2 with Hole



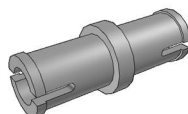
2x
2730 Black
Technic Brick 1 x 10 with Holes



2x
32123a Light Bluish Gray
Technic Bush 1/2 Smooth with Axle



4x
3713 Light Bluish Gray
Technic Bush with Two Flanges



2x
3673 Light Bluish Gray
Technic Pin



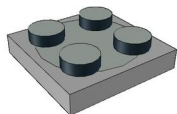
1x
3709b Black
Technic Plate 2 x 4 with Holes



2x
3709b Dark Bluish Gray
Technic Plate 2 x 4 with Holes



1x
4185a Light Bluish Gray
Technic Wedge Belt Wheel - Not Reinforced



2x
3680c01 Light Bluish Gray
Turntable 2 x 2 Plate with Light

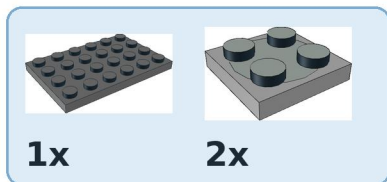


2x
2696 Black
Tyre 12.7/ 51 x 30

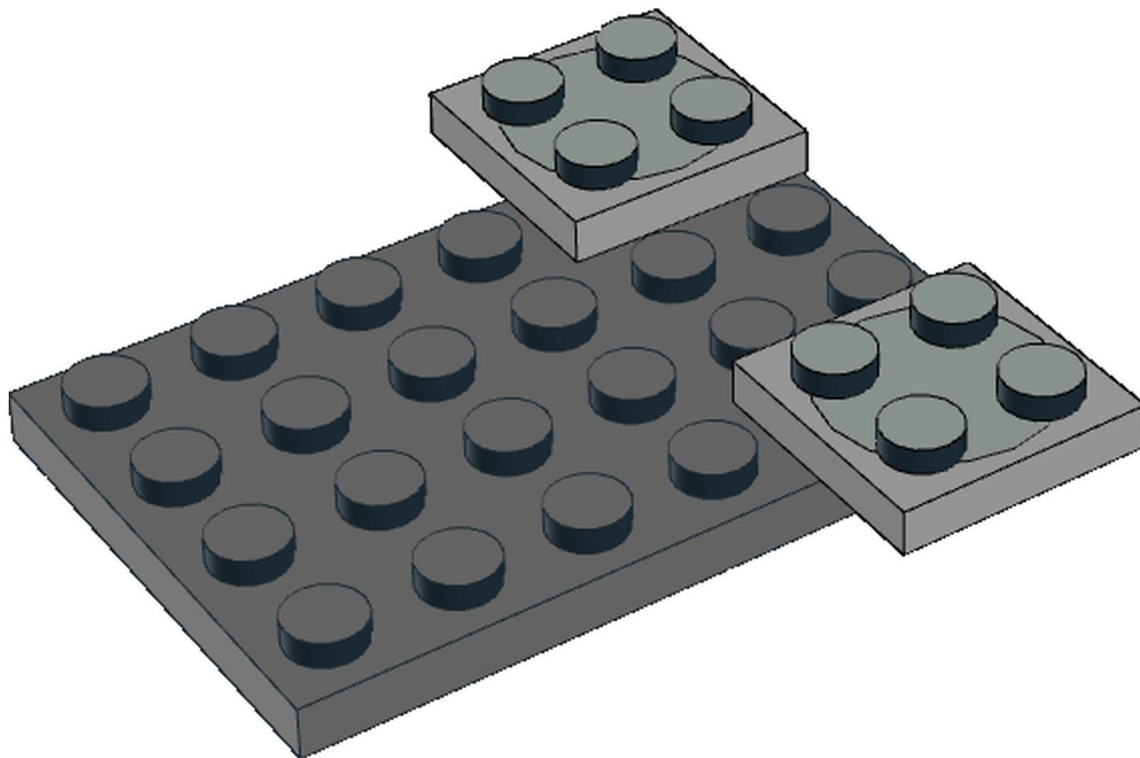


2x
2695 Light Bluish Gray
Wheel Rim 12.7 x 30 Stepped

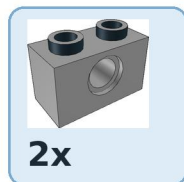
1



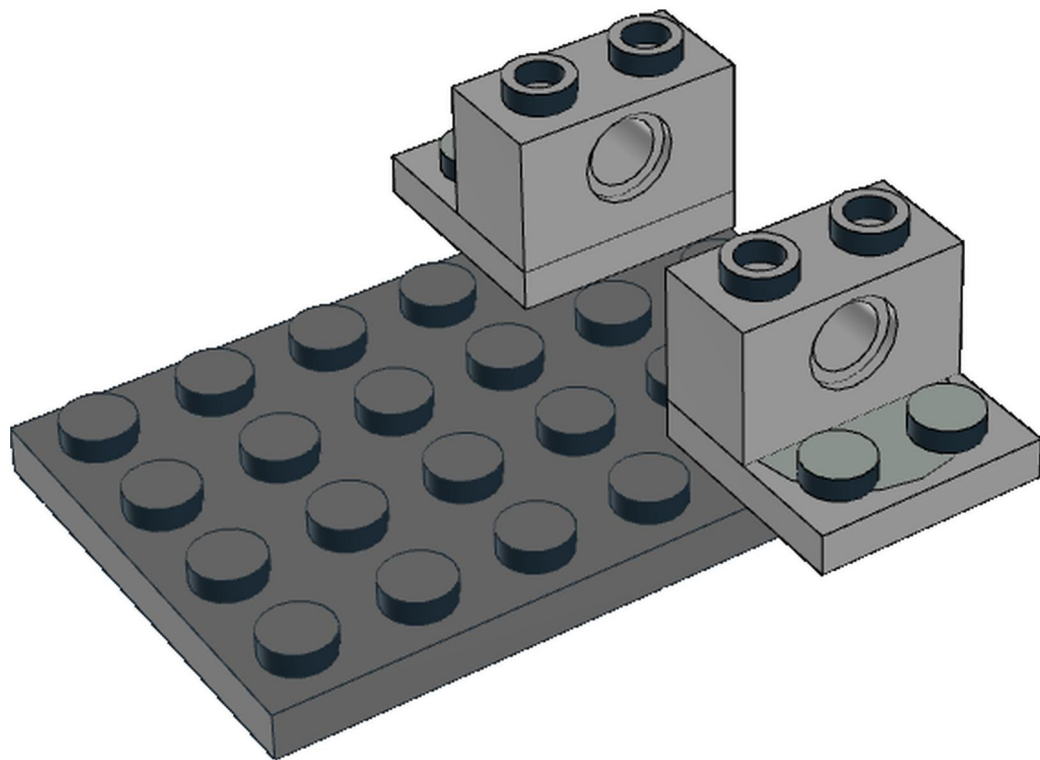
Build the axle beam. The two turntables are the kingpins the front wheels steer on.



2



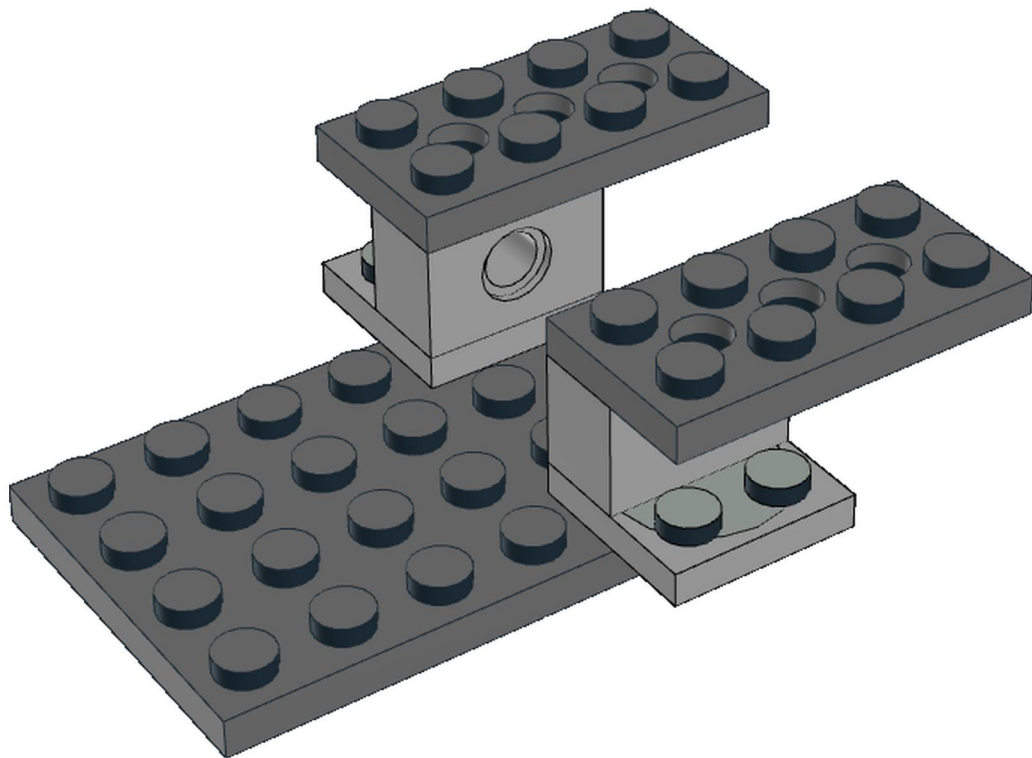
One knuckle on each turntable, holes facing sideways. The knuckle should swivel freely.



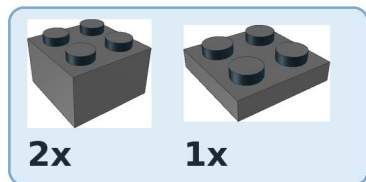
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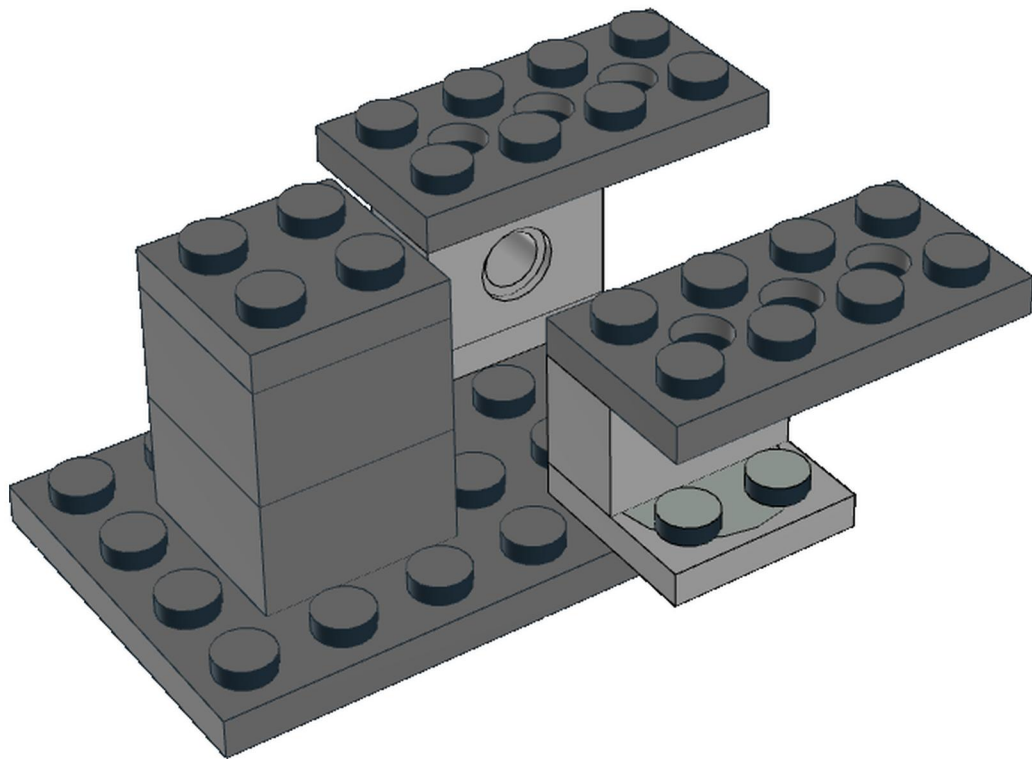
Steering arms: the end holes sit directly over the knuckle; the arm points backwards.



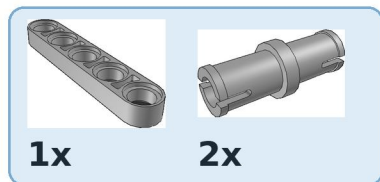
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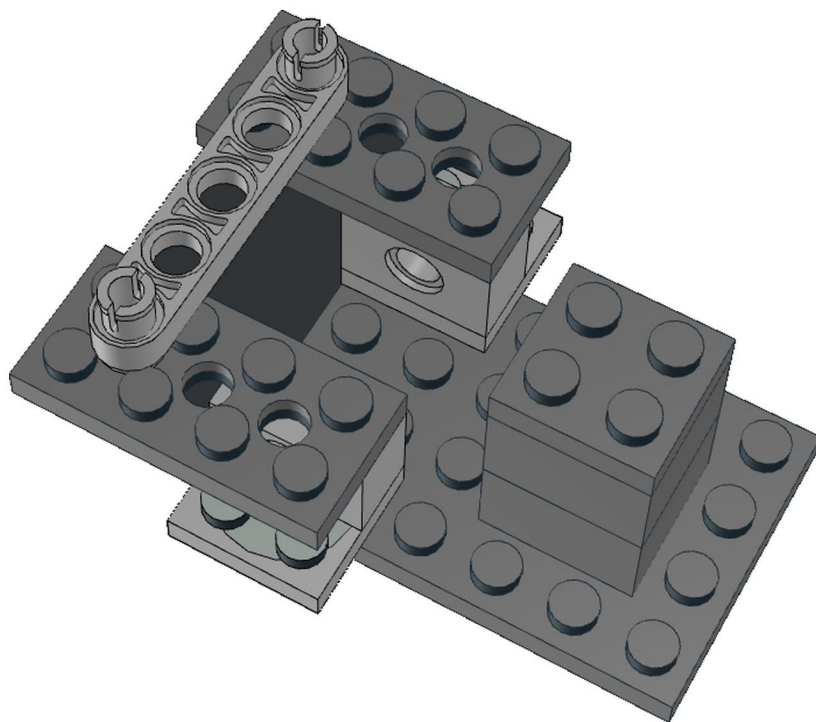
The post in front of the knuckles will carry the frame.



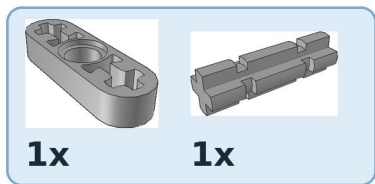
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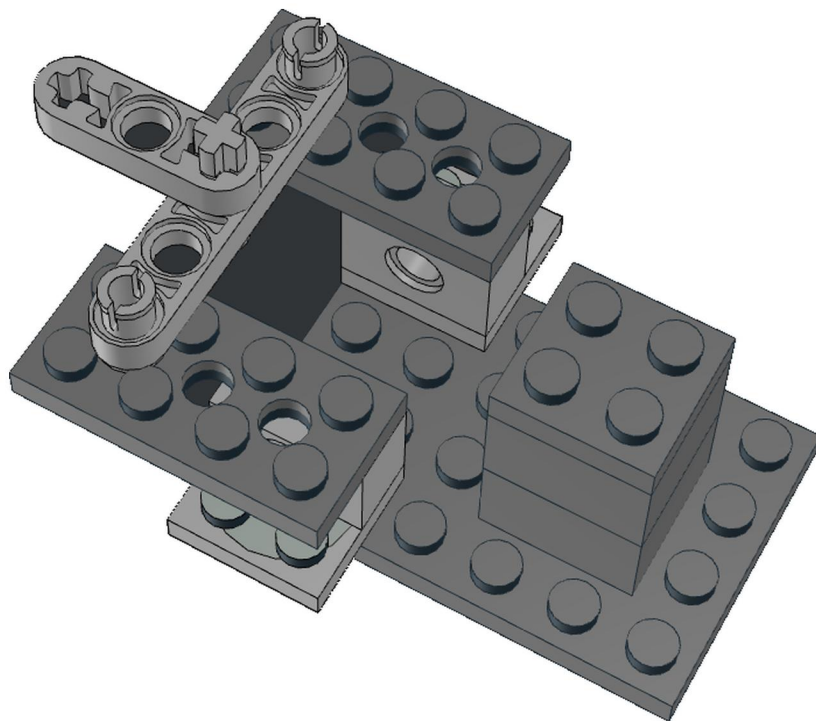
Lay the tie rod across the arm studs (it rests ON the studs, it does not press on).
Push a pin down through each end hole into the steering-arm hole behind the knuckle.



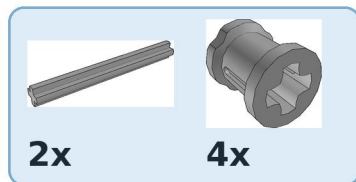
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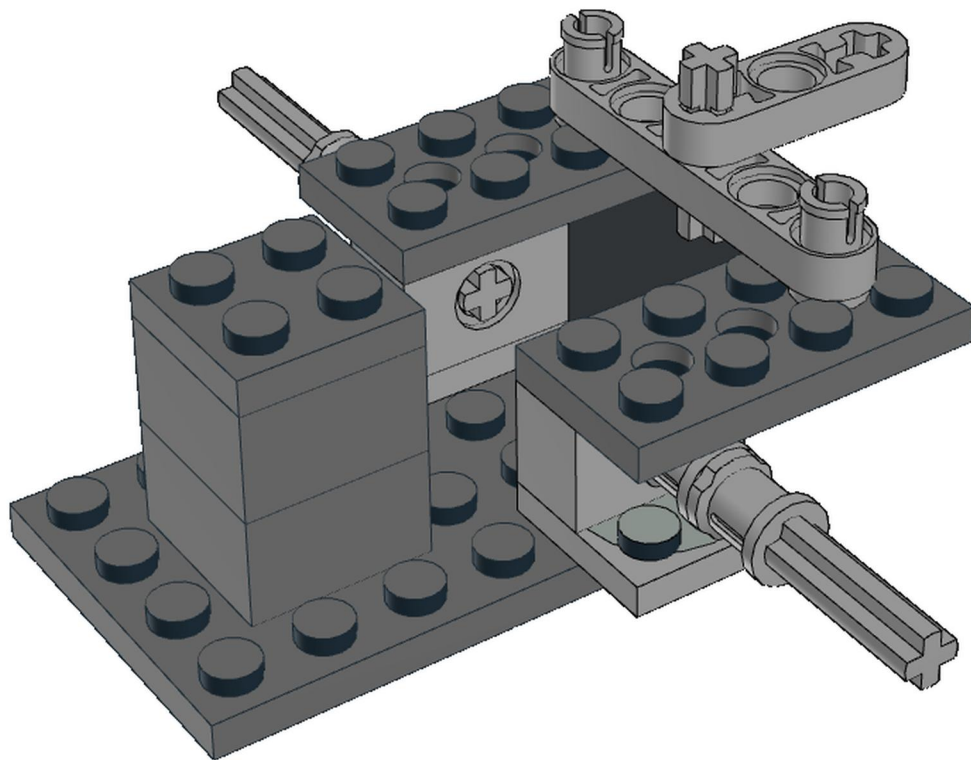
The lever sits on top of the tie rod. The 2L axle goes through the lever's end (axle hole, locked) into the tie rod's centre hole (round, free to turn).



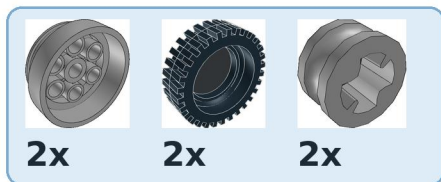
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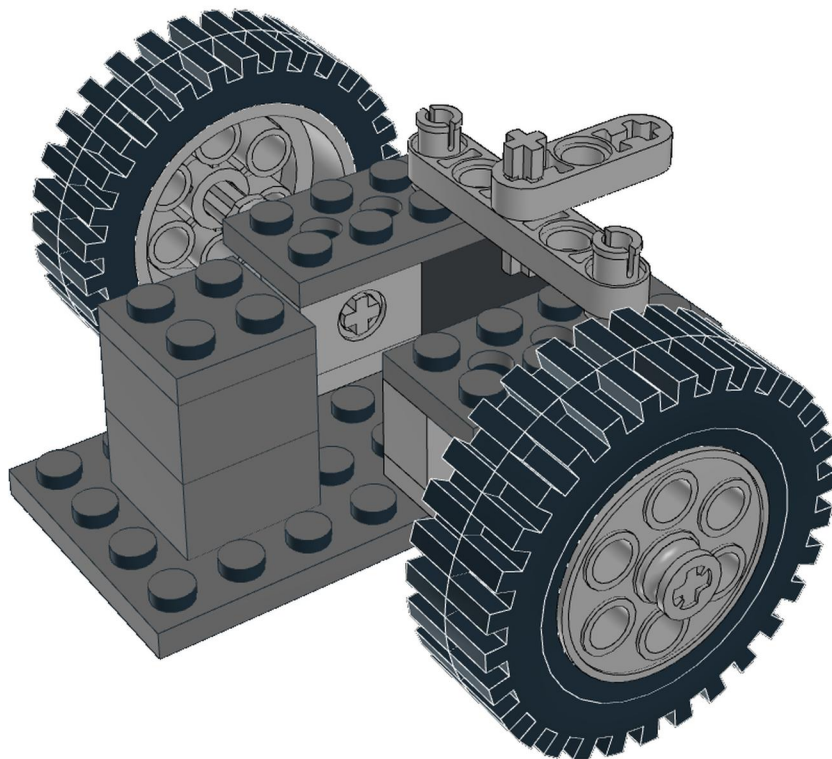
Stub axle through each knuckle, then two bushes on the outside.



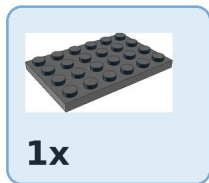
8



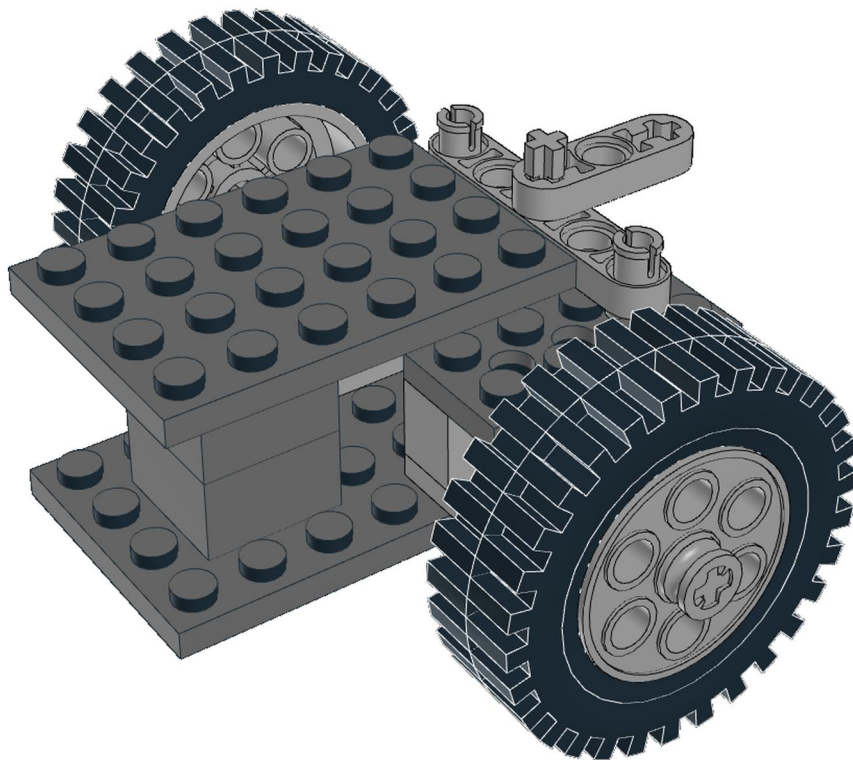
Wheel, then a half bush to hold it on.



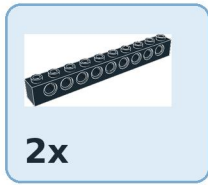
9



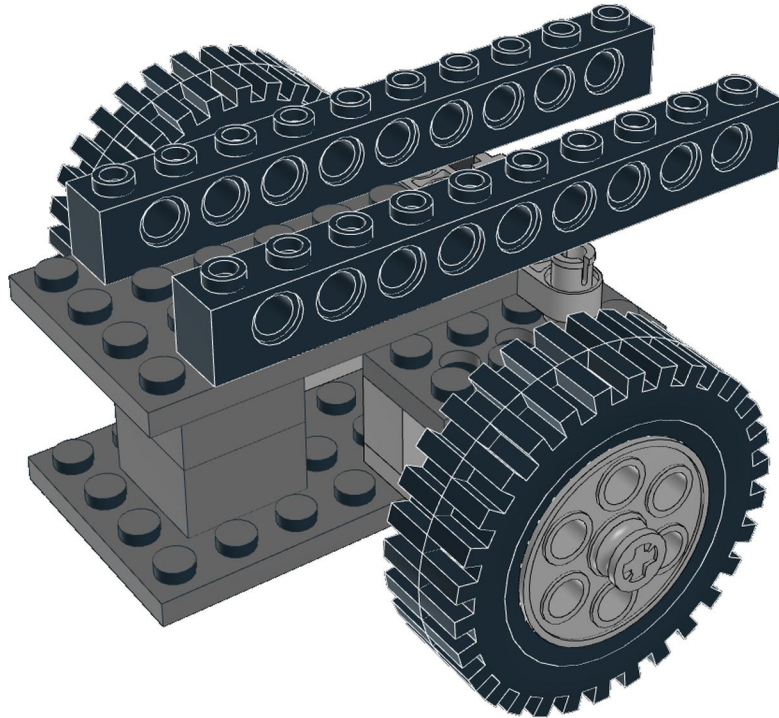
Press the 4 x 6 plate onto the top of the post.



10



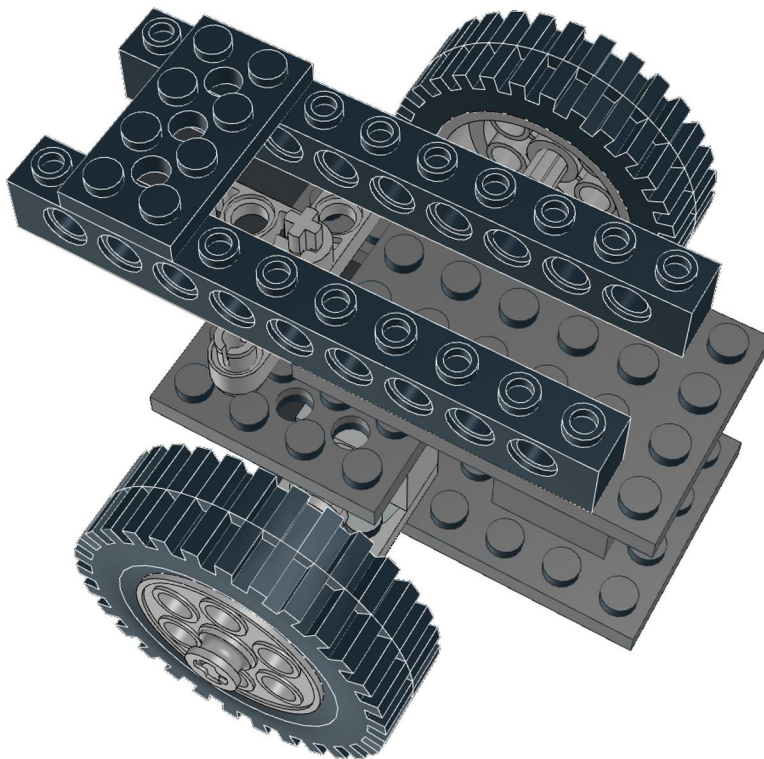
Two frame rails on the plate. Any 1 x 10 bricks work for this test (or 1 x 8 + 1 x 2). Keep them 2 studs apart.



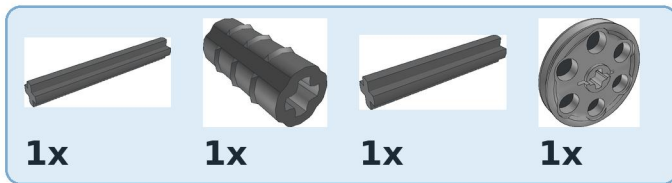
11



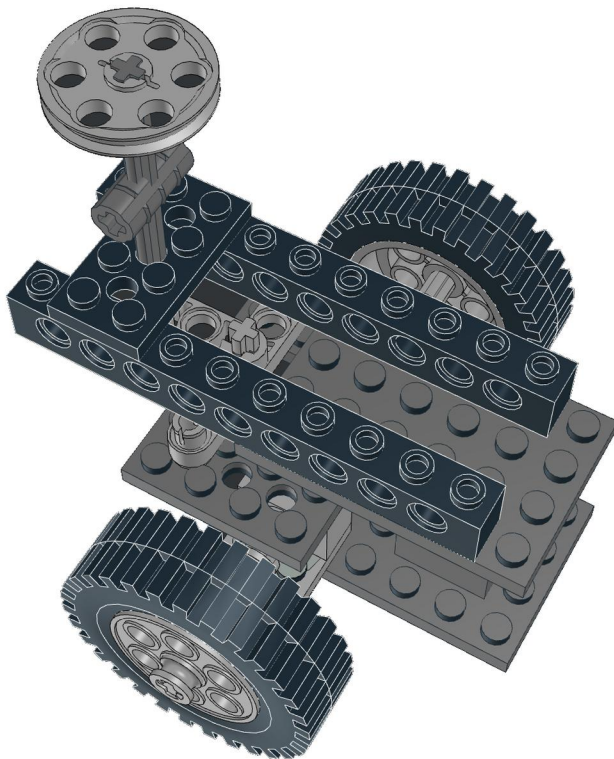
The bearing plate bridges the rails. Its middle hole must be over the lever's free end.



12



Drop the 4L axle through the bearing into the lever. Joiner, 3L axle and knob on top.



Test it

- ☐ Turn the knob slowly lock to lock. Both wheels turn the same way, smoothly, without the tie rod catching on the arm studs.
- ☐ Count the lock: roughly 30-35 degrees each way before a tyre touches the frame rails or the axle beam.
- ☐ Let go at full lock: the wheels should stay put, not spring back.
- ☐ Press down firmly on the frame rails (about the weight of the cab). The turntables should not tilt or wobble; wheels stay upright.
- ☐ Roll it along the table: it runs straight when the knob is centred.
- ☐ Note anything that binds, wobbles or pops off, and how far it steers. A photo from below helps. We fix it before building the cab.